



VIDEO_COURSE

HelixCode Video Course Outline

Course Overview

Course Title: Mastering HelixCode - Distributed AI Development Platform

Duration: 8 hours (16 lessons, 30 minutes each)

Level: Beginner to Advanced

Prerequisites: Basic programming knowledge, familiarity with Go/PostgreSQL

Course Structure

Module 1: Introduction to HelixCode (1 hour)

Lesson 1: Welcome to HelixCode (30 min)

- **Topics:**
 - What is HelixCode?
 - Distributed AI development concepts
 - Use cases and benefits
 - Course roadmap
- **Hands-on:**
 - Installing HelixCode
 - Running first worker
 - Basic configuration
- **Resources:**
 - Course slides
 - Installation guide
 - Quick start checklist

Lesson 2: Architecture Overview (30 min)

- **Topics:**
 - System architecture
 - Distributed worker network
 - MCP protocol integration
 - Security architecture
- **Hands-on:**
 - Exploring project structure

- Understanding configuration files
- Basic system monitoring
- **Resources:**
 - Architecture diagrams
 - Component documentation
 - Security guidelines

Module 2: Core Concepts (2 hours)

Lesson 3: Worker Management (30 min)

- **Topics:**
 - Worker types and capabilities
 - SSH-based worker pools
 - Health monitoring
 - Auto-installation
- **Hands-on:**
 - Adding local workers
 - Configuring SSH workers
 - Worker health checks
- **Resources:**
 - Worker configuration templates
 - SSH setup guide
 - Troubleshooting checklist

Lesson 4: Task Management (30 min)

- **Topics:**
 - Task types and workflows
 - Work preservation features
 - Checkpoint system
 - Task prioritization
- **Hands-on:**
 - Creating basic tasks
 - Monitoring task progress
 - Using checkpoints
- **Resources:**
 - Task templates
 - Work preservation guide
 - Performance optimization tips

Lesson 5: Advanced LLM Integration (30 min)

- **Topics:**

- LLM provider architecture
- Tool calling API
- Reasoning capabilities
- Model management

- **Hands-on:**

- Configuring LLM providers
- Using tool calling
- Implementing reasoning

- **Resources:**

- LLM provider documentation
- Tool integration guide
- Reasoning patterns

Lesson 6: MCP Protocol Integration (30 min)

- **Topics:**

- Model Context Protocol overview
- Tool discovery and execution
- Multi-transport support
- Authentication and security

- **Hands-on:**

- Adding MCP servers
- Using MCP tools
- Custom tool development

- **Resources:**

- MCP specification
- Tool development guide
- Security best practices

Module 3: Development Workflows (2 hours)

Lesson 7: Planning Mode (30 min)

- **Topics:**

- Distributed project analysis
- Technology research
- Architecture design
- Resource planning

- **Hands-on:**

- Creating development plans
- Analyzing project requirements

- Resource allocation
- **Resources:**
 - Planning templates
 - Analysis tools
 - Best practices guide

Lesson 8: Building Mode (30 min)

- **Topics:**
 - Distributed compilation
 - Parallel code generation
 - Build artifact caching
 - Cross-platform builds
- **Hands-on:**
 - Setting up build workflows
 - Optimizing build performance
 - Managing build artifacts
- **Resources:**
 - Build configuration templates
 - Performance optimization guide
 - Caching strategies

Lesson 9: Testing Mode (30 min)

- **Topics:**
 - Distributed test execution
 - Parallel test suites
 - Quality scanning
 - Performance testing
- **Hands-on:**
 - Configuring test environments
 - Running distributed tests
 - Analyzing test results
- **Resources:**
 - Test configuration templates
 - Quality metrics
 - Performance benchmarks

Lesson 10: Refactoring Mode (30 min)

- **Topics:**
 - AI-assisted refactoring
 - Cross-file coordination
 - Safety validation

- Rollback mechanisms
- **Hands-on:**
 - Performing automated refactoring
 - Validating changes
 - Using rollback features
- **Resources:**
 - Refactoring patterns
 - Safety guidelines
 - Validation tools

Module 4: Advanced Features (2 hours)

Lesson 11: Multi-Client Support (30 min)

- **Topics:**
 - REST API development
 - Terminal UI implementation
 - CLI interface
 - Mobile applications
- **Hands-on:**
 - Using REST API
 - Customizing terminal UI
 - Building CLI tools
- **Resources:**
 - API documentation
 - UI customization guide
 - CLI development kit

Lesson 12: Notification System (30 min)

- **Topics:**
 - Multi-channel notifications
 - Rule-based routing
 - Template system
 - Priority management
- **Hands-on:**
 - Configuring notification channels
 - Creating notification rules
 - Customizing templates
- **Resources:**
 - Notification templates
 - Channel configuration guide
 - Routing rules examples

Lesson 13: Security and Authentication (30 min)

- **Topics:**
 - JWT authentication
 - Role-based access control
 - Encryption
 - Audit logging
- **Hands-on:**
 - Setting up authentication
 - Configuring access controls
 - Implementing encryption
- **Resources:**
 - Security configuration guide
 - Access control templates
 - Audit logging setup

Lesson 14: Performance Optimization (30 min)

- **Topics:**
 - System monitoring
 - Resource optimization
 - Caching strategies
 - Load balancing
- **Hands-on:**
 - Setting up monitoring
 - Optimizing resource usage
 - Implementing caching
- **Resources:**
 - Performance monitoring tools
 - Optimization checklist
 - Caching configuration guide

Module 5: Production Deployment (1 hour)

Lesson 15: Deployment Strategies (30 min)

- **Topics:**
 - Docker deployment
 - Kubernetes orchestration
 - Cloud deployment
 - High availability
- **Hands-on:**
 - Deploying with Docker Compose
 - Kubernetes configuration

- Cloud deployment setup
- **Resources:**
 - Deployment scripts
 - Kubernetes manifests
 - Cloud configuration templates

Lesson 16: Monitoring and Maintenance (30 min)

- **Topics:**
 - System monitoring
 - Log management
 - Backup strategies
 - Troubleshooting
- **Hands-on:**
 - Setting up monitoring
 - Configuring logging
 - Implementing backups
- **Resources:**
 - Monitoring configuration
 - Log management guide
 - Backup scripts

Course Resources

Video Production Requirements

Equipment:

- **Camera:** 4K capable DSLR/mirrorless
- **Audio:** Lavalier microphone + studio recording
- **Lighting:** Three-point lighting setup
- **Software:** Final Cut Pro / Adobe Premiere

Production Schedule:

- **Pre-production:** 2 weeks
- **Recording:** 3 weeks
- **Post-production:** 3 weeks
- **Review and QA:** 1 week

Course Materials

Student Workbook:

- Step-by-step exercises
- Configuration templates
- Troubleshooting guides
- Best practices checklist

Code Repository:

- Complete project examples
- Configuration files
- Deployment scripts
- Test suites

Assessment:

- Quizzes after each module
- Hands-on projects
- Final certification exam

Marketing and Distribution

Target Audience:

- Software developers
- DevOps engineers
- AI/ML engineers
- System administrators

Distribution Platforms:

- Udemy
- Coursera
- YouTube (free introductory content)
- Corporate training programs

Pricing Strategy:

- **Individual:** \$199
- **Team (5+):** \$149 per person
- **Enterprise:** Custom pricing
- **Free:** First module + certification exam

Success Metrics

Course Completion:

- **Target:** 80% completion rate
- **Engagement:** 90% video watch time
- **Satisfaction:** 4.5+ star rating

Business Impact:

- **Revenue:** \$50,000 in first 6 months
- **Students:** 500+ enrolled in first year
- **Retention:** 70% course completion

Community Building:

- **Forum:** Active student community
- **Contributions:** Open source contributions
- **Partnerships:** Industry collaborations

Course Development Timeline: 12 weeks

Production Budget: \$15,000

Expected ROI: 300% in first year

Support: Dedicated course support team